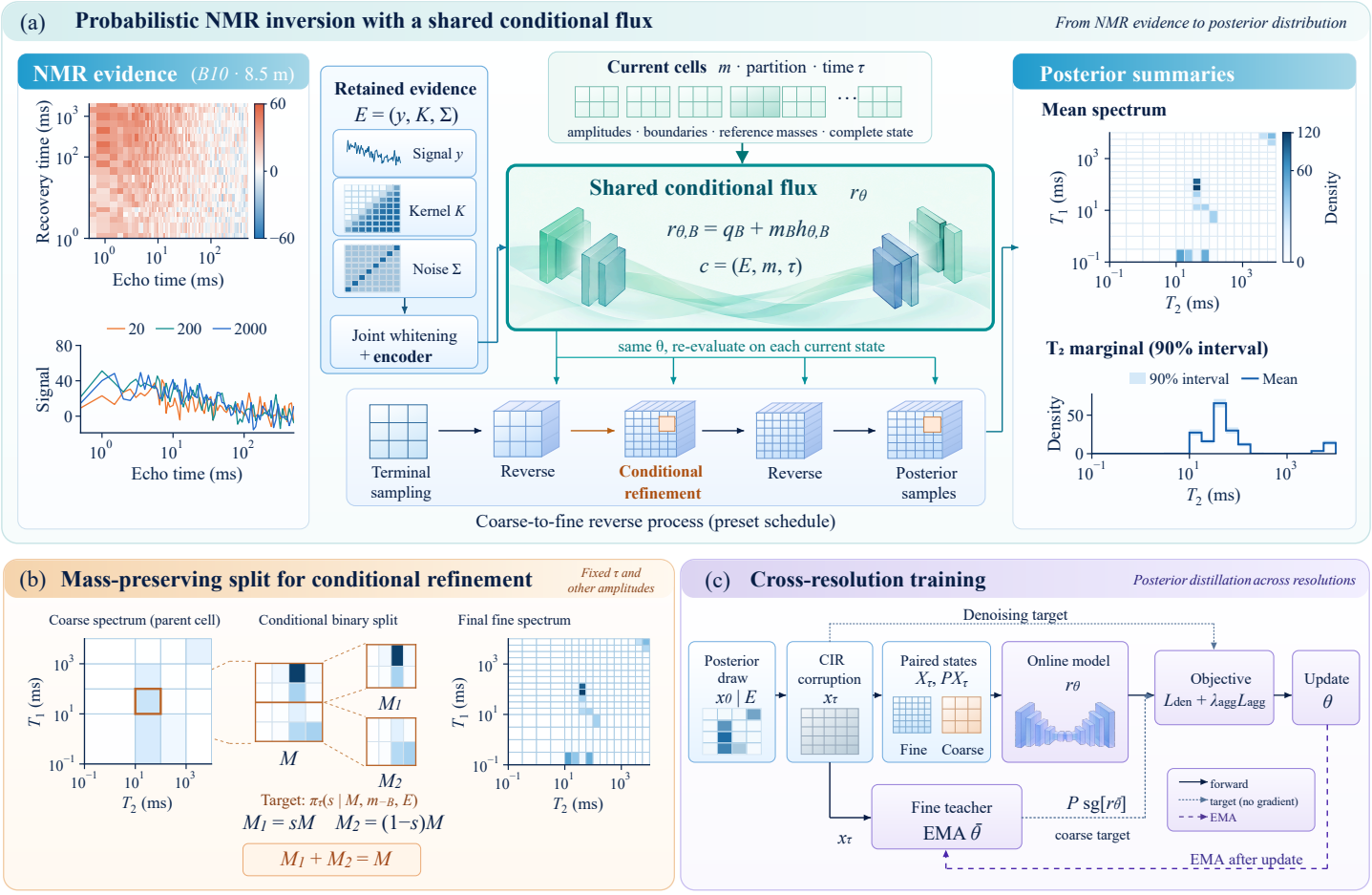

(a) Probabilistic NMR inversion with a shared conditional flux
From NMR evidence to posterior distribution
NMR evidence (B10 · 8.5 m)
Recovery time (ms)
Echo time (ms)
60
0
−60
20
200
2000
Signal
Echo time (ms)
Retained evidence
E = (y, K, Σ)
Signal y
Kernel K
Noise Σ
Joint whitening + encoder
Current cells m · partition · time τ
amplitudes · boundaries · reference masses · complete state
Shared conditional flux rθ
rθ,B = qB + mBhθ,B
c = (E, m, τ)
same θ, re-evaluate on each current state
Terminal sampling
Reverse
Conditional refinement
Reverse
Posterior samples
Coarse-to-fine reverse process (preset schedule)
Posterior summaries
Mean spectrum
T1 (ms)
T2 (ms)
Density
120
60
0
T₂ marginal (90% interval)
90% interval
Mean
Density
50
0
T2 (ms)
(b) Mass-preserving split for conditional refinement
Fixed τ and other amplitudes
Coarse spectrum (parent cell)
Conditional binary split
Final fine spectrum
M
M1
M2
T1 (ms)
T2 (ms)
Target: πτ(s | M, m−B, E)
M1 = sM M2 = (1−s)M
M1 + M2 = M
(c) Cross-resolution training
Posterior distillation across resolutions
Denoising target
Posterior draw x0 | E
CIR corruption xτ
Paired states Xτ, PXτ
Fine
Coarse
Online model rθ
Objective Lden + λaggLagg
Update θ
Fine teacher EMA θ̄
P sg[rθ̄]
coarse target
xτ
forward
target (no gradient)
EMA
EMA after update